animals in last fifteen years, such as wild boar and tiger. There are multiple reasons for their poaching and hunting, such as their valuable organs like tusk, bone, and teeth. At times animals attack because of their health issues, environmental changes, and food unavailability issues.

Hence, constant monitoring and surveillance of the wildlife is important to protect them. Conservation drones and modern technologies help in designing a robust solution to handle this efficiently. Using these drones, we can do surveillance remotely and understand the lifestyle of animals and help them to manage life better.

According to a research report from the studies published in *wwf.org.uk*, the fourth largest illegal activity across globe is wildlife poaching. (First three are drug trafficing, child/women trafficking, and counterfeiting.) Modern drone solutions like ideaForge Drones help develop an AI-driven solution for image pattern matching where the endurance cameras capture images continuously. The images are transmitted to remote cloud servers and analysed using AI solutions to understand the anomaly in wildlife movement. This helps in designing a counter poaching approach like alarm signals, diversions using sound notifications, and sensor based protection like dynamic electric fence.

The most popular anti-poaching solutions enabled and equipped by drone systems, which are popular for research and industrial usage, are listed below:

- 100km Skywalker
- Air Shepherd ZT-TIC
- DJI Inspire 1
- Sensefly eBee Plus
- Silent Falcon
- Super Bat Da-50
- Zeta FX-61 Phantom

Such anti-poaching solutions are specially designed for wildlife monitoring and are powered by AI-enabled customised software applications (as bundled license). These can be used for wildlife research related activities like census, animal tracking and migration tracing, habitat management for wildlife in seasonal changes, nest monitoring during incubation, and species identification.

Physical Specifications:
- Weight: 2.6kg
- Dimensions: 30×20×15cm
- Nominal battery life: 2-3 hours

Performance Specifications:
- Maximum depth: 75m (we have taken them past 100m, but true depth depends on build quality)
- Maximum tether length: 300m (100m tether provided)
- Maximum forward speed: 2 knots
- LED brightness: 180 lm

Key Features:
- Live HD video with wide-angle lens and tilt function
- LED lighting for low-light environments
- Easy-to-use Web browser user interface
- Beagle bone Black AND Arduino MEGA microprocessors for a flexible and powerful developer platform with dozens of input/output channels and plenty of computing power for user-designed features and experiments
- Rechargeable lithium batteries (sold separately)
- 100m lightweight 2-wire tether
- Compact and lightweight backpack or carry-on ready
- Payload area for additional hardware or equipment

Fig. 4: OpenROV

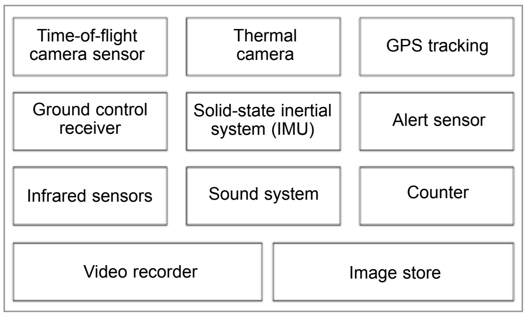

Fig. 5: Conservation drone system for wildlife monitoring

Marine and land mapping

Geospatial information system based drone services are very helpful for both forest and marine (sea/ocean) level monitoring and mapping. These help to understand and study the landscapes, the density of animals and wildlife movements in different zones, climatic changes and their impact on the lifestyle of wildlife and, above all, tracking the poaching activities and devise an anti-poaching solution.

Radio telemetry can be used for animal tracking in forest and marine species monitoring in sea/ocean. For radio tracking animals, they must be tagged with VHF-radio tags to track from remote drone sensors to know their movement and count the animals.

Image analysis software like Agisoft and ImageJ, along with machine learning technique for wildlife availability, can be very useful for data analysis of thermal images captured from drones and used for animal counting and tracking.

These applications are popular in Britain, Germany, and Australia in dense forests and marine areas for wildlife monitoring and tracking. They reduce the poaching activities to a great extent. EFY